

Lecture 20

CLT

Let $X_1, X_2, \dots, X_n$ be iid with mean μ and var $\sigma^2 > 0$. Then,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Remark. Typically, one writes:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

However, this is exactly the same as

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

It is because: We know $\forall t \in \mathbb{R}$:

$$P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq t\right) \xrightarrow{n \rightarrow \infty} \int_{-\infty}^t \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

Hence $\forall t \in \mathbb{R}$:

$$\Rightarrow P(\sqrt{n}(\bar{X}_n - \mu) \leq \sigma t) \xrightarrow{n \rightarrow \infty} \int_{-\infty}^t \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

Let $z = \sigma t \Leftrightarrow t = z/\sigma$.

$$\Rightarrow P(\sqrt{n}(\bar{X}_n - \mu) \leq z) \xrightarrow{n \rightarrow \infty} \int_{-\infty}^{z/\sigma} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

Let $y = \sigma x \Rightarrow dx = \frac{dy}{\sigma}$.

$$= \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2\sigma^2}} \frac{dy}{\sigma}$$

This is $P(Z \leq z)$ where $Z \sim N(0, \sigma^2)$.

$$\Rightarrow \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Often your statistic need not be $\bar{X}_n$, but a functional transform of such!

Example. Assume we are doing a success/failure trial with results $X_1, X_2, \dots, X_n$ iid Bernoulli(p).

Statistical Notion:

To find p is "unknown", but we know $0 < p < 1$. A "good" estimator for p is $\hat{p} = \bar{X}_n$. Let $S_n = \sum_{i=1}^n X_i \sim \text{Bin}(n, p)$.

CLT (De-Moivre Laplace CLT)

$$\frac{S_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0, 1)$$

This is the "Normal Approximation to the Binomial Dist."

$$\begin{aligned} P(a < S_n \leq b) &= P\left(\frac{a - np}{\sqrt{np(1-p)}} < \frac{S_n - np}{\sqrt{np(1-p)}} \leq \frac{b - np}{\sqrt{np(1-p)}}\right) \\ &\approx \int_{a_n}^{b_n} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \end{aligned}$$

(where a_n and b_n are the standardized bounds)

Note:

1. $\hat{p}$ is unbiased, $E[\hat{p}] = p \forall 0 < p < 1$.
2. WLLN: $\hat{p} \xrightarrow{P} p$ as $n \rightarrow \infty$.
3. SLLN: $\hat{p} \xrightarrow{a.s.} p$ as $n \rightarrow \infty$.

However, often for statisticians, $\hat{p}$ is of no interest, but $T = -\log(\hat{p})$ is of more interest!

- WLLN: $T \xrightarrow{P} -\log p$ as $n \rightarrow \infty$.
- SLLN: $T \xrightarrow{a.s.} -\log p$ as $n \rightarrow \infty$.

However what about the asymptotic error of $(T - (-\log p))$? Is there a distributional limit of $(T - (-\log p))$ after an appropriate centering & scaling?

Notice: $T = -\log \hat{p} = -\log \frac{S_n}{n}$

$$= \log n - \log S_n = \log n - \log \left(\sum_{i=1}^n X_i \right)$$

Any CLT applicable?

Suppose $g : \mathbb{R} \rightarrow \mathbb{R}$. Consider $T_n = g(\bar{X}_n)$, where $X_1, \dots, X_n$ are iid with mean μ and var $\sigma^2 > 0$. Note:

1. We may not say that $E[T] = g(\mu)$.
2. WLLN: $g(\bar{X}_n) \xrightarrow{P} g(\mu)$ if g is continuous at μ .
3. SLLN: $g(\bar{X}_n) \xrightarrow{a.s.} g(\mu)$ if g is continuous at μ .

δ -Method (of statistics)

Theorem 1. Let $X_1, X_2, \dots, X_n$ be iid with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function which is differentiable and $g' : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at μ . Then:

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2)$$

We get a new asymptotic variance $\sigma^2(g'(\mu))^2$.

Example continued

$X_1, X_2, \dots, X_n$ iid Bern(p). $\hat{p}_n = \bar{X}_n$.

We know $\sqrt{n}(\bar{X}_n - p) \xrightarrow{d} N(0, p(1-p))$.

Here, $\mu = p$ and $\sigma^2 = p(1-p)$.

Let $T_n = -\log \hat{p}_n$. We use $g(x) = -\log x$.

$\forall x \in (0, 1)$ Then $g'(x) = -\frac{1}{x}$.

By the Delta-Method Theorem:

$$\sqrt{n}(T_n - (-\log p)) \xrightarrow{d} N(0, p(1-p)(g'(p))^2)$$

$$\sqrt{n}(T_n + \log p) \xrightarrow{d} N\left(0, p(1-p)\left(-\frac{1}{p}\right)^2\right)$$

$$\sqrt{n}(T_n + \log p) \xrightarrow{d} N\left(0, \frac{1-p}{p}\right)$$

Variance Stabilizing Transform

We observe T_n has an asymptotic variance that depends on p . (The variance of $\bar{X}_n$ is $\frac{p(1-p)}{n}$).

Can we find a transform g s.t. $g(\bar{X}_n)$ will have the asymptotic variance free of p ? The asymptotic variance of $\sqrt{n}(g(\bar{X}_n) - g(p))$ is $\sigma^2(g'(p))^2 = p(1-p)(g'(p))^2$. We want this to be constant, e.g., 1.

$$p(1-p)(g'(p))^2 = 1$$

$$g'(p) = \frac{1}{\sqrt{p(1-p)}} \quad (0 < p < 1)$$

To find $g(p)$, we integrate:

$$\int \frac{dp}{\sqrt{p(1-p)}} = \arcsin p = \sin^{-1} p$$

Then,

$$\sqrt{n}(\sin^{-1}(\bar{X}_n) - \sin^{-1}(p)) \xrightarrow{d} N(0, 1)$$

This is called a "Variance Stabilizing Transform". If σ is a function of μ then, there exists a function g s.t. taking $g(x) = \int \frac{1}{\sigma(x)} dx$ will have asymptotic variance parameter free of variables i.e. constant.

Proof of δ -Method

By CLT:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Assume $g : \mathbb{R} \rightarrow \mathbb{R}$ is cont. & differentiable with g' cont. at μ . By Mean Value Theorem:

$$g(\bar{X}_n) - g(\mu) = g'(\xi_n)(\bar{X}_n - \mu)$$

where ξ_n is some value between $\bar{X}_n$ and μ , so $|\xi_n - \mu| < |\bar{X}_n - \mu|$.

By WLLN: $\bar{X}_n \xrightarrow{P} \mu$. Since ξ_n is squeezed between $\bar{X}_n$ and μ , we also have $\xi_n \xrightarrow{P} \mu$. Since g' is continuous at μ :

$$g'(\xi_n) \xrightarrow{P} g'(\mu) \quad (\text{by Continuous Mapping Theorem})$$

Now consider the full expression:

$$\begin{aligned} \sqrt{n}(g(\bar{X}_n) - g(\mu)) &= \sqrt{n}(g'(\xi_n)(\bar{X}_n - \mu)) \\ &= g'(\xi_n) \cdot [\sqrt{n}(\bar{X}_n - \mu)] \end{aligned}$$

We have:

- $g'(\xi_n) \xrightarrow{P} g'(\mu)$ (a constant)
- $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$ (a random variable)

By Slutsky's Theorem:

$$g'(\xi_n) \cdot [\sqrt{n}(\bar{X}_n - \mu)] \xrightarrow{d} g'(\mu) \cdot N(0, \sigma^2)$$

Which is:

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2)$$

Proof of δ -Method (continued)

Assume $E[Z_n] = \mu$

$$\sqrt{n}(Z_n - E[Z_n]) \xrightarrow{d} N(0, \sigma^2)$$

$g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous & differentiable with g' continuous at μ .

By Mean Value Theorem:

$$g(Z_n) - g(\mu) = g'(\xi_n)(Z_n - \mu)$$

where ξ_n is some value b/w Z_n & μ .

$$\begin{aligned} |\xi_n - \mu| &< |Z_n - \mu| \\ \Rightarrow \quad \xi_n &\xrightarrow{P} \mu \quad \Rightarrow \quad g'(\xi_n) \xrightarrow[n \rightarrow \infty]{P} g'(\mu) \end{aligned}$$

Assumption:

$$\begin{aligned} \sqrt{n}(Z_n - \mu) &\xrightarrow{d} N(0, \sigma^2) \\ \Rightarrow (Z_n - \mu) &= \frac{\sqrt{n}(Z_n - \mu)}{\sqrt{n}} \xrightarrow{d} 0 \end{aligned}$$

$$X_n \xrightarrow{d} c \iff X_n \xrightarrow{P} c$$

$\Rightarrow$ By Slutsky's Theorem:

$$\sqrt{n}(g(Z_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2)$$

General δ -method

Suppose T_n is a statistic, s.t. $\exists \mu \in \mathbb{R}$ and $\sigma^2 > 0$ constant with

$$\sqrt{n}(T_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable & g' is continuous at μ . Then

$$\sqrt{n}(g(T_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2)$$

An example is CLT

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

$X_1, \dots, X_n$ iid mean μ & var $\sigma^2 > 0$

L-20 ends